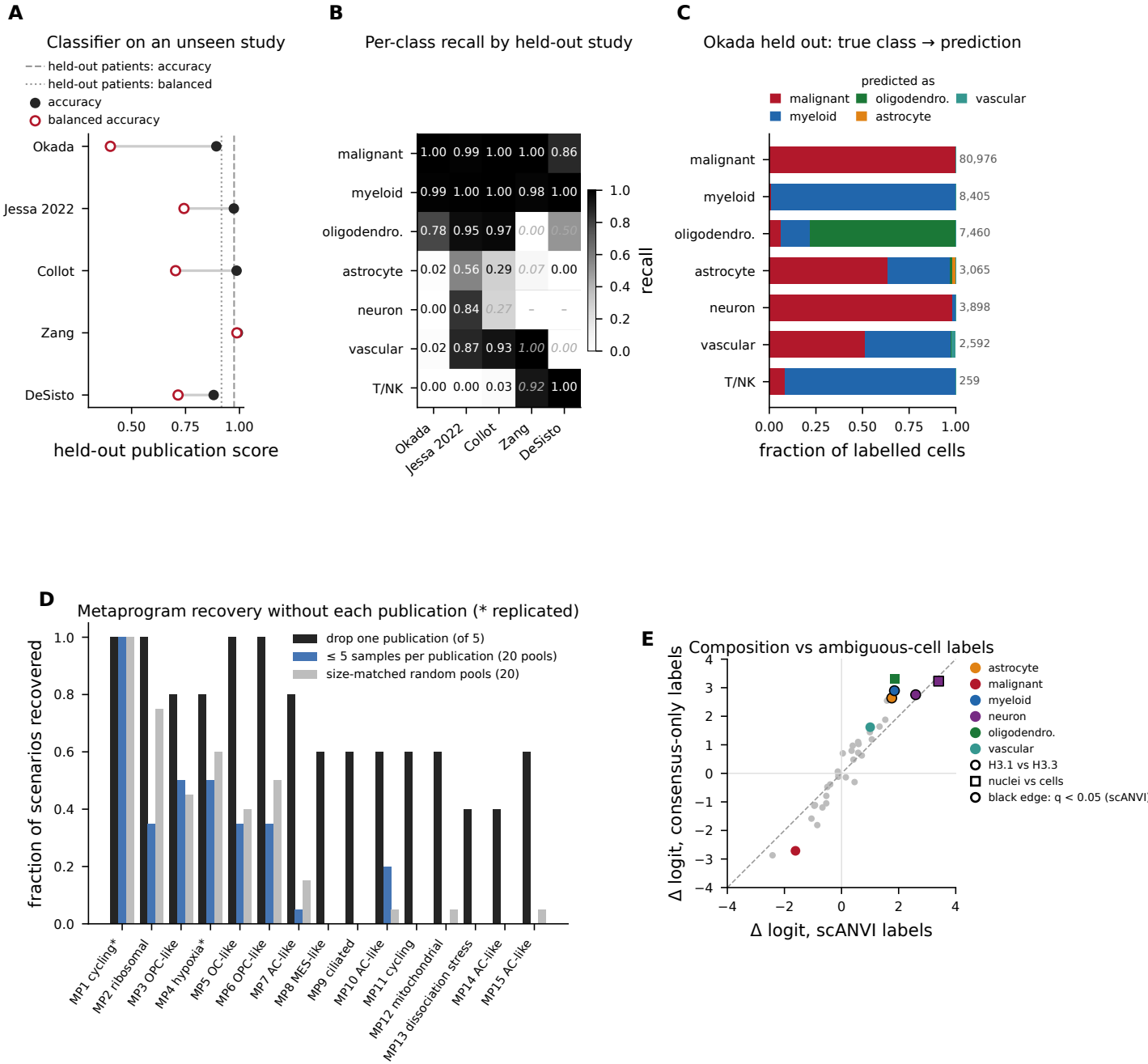
A
Classifier on an unseen study
held-out patients: accuracy
held-out patients: balanced
accuracy
balanced accuracy
Okada
Jessa 2022
Collot
Zang
DeSisto
0.50
0.75
1.00
held-out publication score
B
Per-class recall by held-out study
malignant 1.00 0.99 1.00 1.00 0.86
myeloid 0.99 1.00 1.00 0.98 1.00
oligodendro. 0.78 0.95 0.97 0.00 0.50
astrocyte 0.02 0.56 0.29 0.07 0.00
neuron 0.00 0.84 0.27 – –
vascular 0.02 0.87 0.93 1.00 0.00
T/NK 0.00 0.00 0.03 0.92 1.00
Okada
Jessa 2022
Collot
Zang
DeSisto
recall
1.0
0.8
0.6
0.4
0.2
0.0
C
Okada held out: true class → prediction
predicted as
malignant
oligodendro.
vascular
myeloid
astrocyte
malignant 80,976
myeloid 8,405
oligodendro. 7,460
astrocyte 3,065
neuron 3,898
vascular 2,592
T/NK 259
0.00
0.25
0.50
0.75
1.00
fraction of labelled cells
D
Metaprogram recovery without each publication (* replicated)
drop one publication (of 5)
≤ 5 samples per publication (20 pools)
size-matched random pools (20)
fraction of scenarios recovered
1.0
0.8
0.6
0.4
0.2
0.0
MP1 cycling*
MP2 ribosomal
MP3 OPC-like
MP4 hypoxia*
MP5 OC-like
MP6 OPC-like
MP7 AC-like
MP8 MES-like
MP9 ciliated
MP10 AC-like
MP11 cycling
MP12 mitochondrial
MP13 dissociation stress
MP14 AC-like
MP15 AC-like
E
Composition vs ambiguous-cell labels
Δ logit, consensus-only labels
Δ logit, scANVI labels
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
astrocyte
malignant
myeloid
neuron
oligodendro.
vascular
H3.1 vs H3.3
nuclei vs cells
black edge: q < 0.05 (scANVI)